

Sessional Exam 1
Introduction to Data Science and Design Thinking (BCE-O633)

Note: Section A contains 4 questions attempt any 2 (5 marks Each), Section B contains 2 questions, attempt any 1 (10 marks each)

Section A

- Q1. Define the term Data Science, and explain steps to learn data science.
- Q2. Discuss the 5 v's in terms of data.
- Q3. What are the stages in a data science project? explain each phase in brief.
- Q4. What are data security issues in a data science project?

Section B

- Q1. You own a bike share company and you want to increase your revenue in next semester. What will be your policy to achieve the goal? Discuss your approach with respect to different Data Science Roles and explain the task of each persona.
- Q2. What are the applications of Data Science in various fields? Mention any five applications with detail.

Sessional Exam 2
Introduction to Data Science and Design Thinking (BCE-O633)

Note: Section A contains 4 questions attempt any 2 (5 marks Each), Section B contains 2 questions, attempt any 1 (10 marks each)

Section A

- Q1. Define Data Integration and its challenges.
- Q2. Discuss data collection strategies. Mention any five data collection methods.
- Q3. Find mean, median and mode for the following data
- 34, 45, 23, 87, 96, 34, 56, 78, 23, 11, 22, 23, 101, 124
- Q4. Create a box plot for the above data. Find all five elements of box plot.

Section B

- Q1. Define data pre-processing and its major task and also discuss why it is required.
- Q2. Discuss data transformation strategies, define any five strategies in detail.

BCE-O633
SEMESTER EXAMINATION-2023
CLASS – B.TECH SEMESTER VI
INTRODUCTION TO DATA SCIENCE AND DESIGN THINKING

Time: 3 hour

Max. Marks: 70

Note: Question Paper is divided into two sections: **A and B**. Attempt both the sections as per given instructions.

SECTION-A (SHORT ANSWER TYPE QUESTIONS)

Instructions: Answer any five questions in about 150 words each. Each question carries six marks.
(5 X 6 = 30 Marks)

Question-1: Explain the term Data Science and discuss its application.

Question-2: Discuss the Data Science Roles in development of Data Science projects.

Question-3: What are the 5 properties of data with respect to Data Science.

Question-4: Define Data transformation and its challenges.

Question-5: Define Data Pre -Processing? Why it is required?

Question-6: Elaborate the process of Data Reduction.

Question-7: Discuss the concept of Empathy, Sympathy and communication in design thinking

Question-8: Discuss the properties of a good design. What are the differences of good design and bad design?

Question-9: What do you mean by Heat map? Where it is used.

Question-10: How to incorporate the design thinking principles in data science? Discuss

SECTION-B (LONG ANSWER TYPE QUESTIONS)

Instructions: Answer any FOUR questions in detail. Each question carries 10 marks.
(4 X 10 = 40 Marks)

Question-1: What are the Data Security Issues? Write any 5 data security issues in detail.

Question-2: What is Data Collection? What are the common methods of data collection?

Question-3: What are the stages in a data science project and how it maps with data analytics life cycle

Question-4: Find mean, median and mode of the following data and then create a box plot.

24, 22, 34, 112, 12, 145, 56, 78, 90, 34, 21, 67, 90, 55, 23, 50, 34

Question-5: What is Data Integration? What are the challenges of data integration?

Question-6: Discuss the stages of design thinking process? Explain each stage in detail with example.

Question-7: Discuss various applications of Design thinking for Rural Development.

Question-8: What is the importance of Nature inspired designing? Discuss the examples of design that are inspired from nature.

Sessional Exam I

Date: 19/02/2025

Time: 2:00- 3:00 PM

Introduction to Data Science and Design Thinking (BCE-O633)

Note: Section A contains 4 questions of 6 marks each attempt any 2.

Section B contains 2 questions of 8 marks each, attempt any 1.

Section A

~~Q1.~~ Define the term Data Science, and explain steps to learn data science.

CO1

~~Q2.~~ Discuss the 5 characteristics of data

CO1

Q3. What are the stages in a data science project? explain each phase in brief.

CO2

Q4. What are data security issues in a data science project?

CO2

Section B

Q1. You own a bike share company and you want to increase your revenue in next semester. What will be your policy to achieve the goal? Discuss your approach with respect to different Data Science Roles and explain the task of each persona.

CO1

~~Q2.~~ What are the applications of Data Science in various fields? Mention any five applications with detail.

CO1

Introduction to Data Science and Design Thinking (BCE-O633)

Sessional Exam II

Date: 03/04/2025

Time: 2:00- 3:00 PM

Note: Section A contains 4 questions of 6 marks each attempt any 2.

Section B contains 2 questions of 8 marks each, attempt any 1.

Section A

S.No	Question	CO	BL
Q1.	What is data Collection? Discuss the different methods for primary Data Collection.	CO3	3
Q2.	What is Data Reduction? Discuss any two Data Reduction Strategies.	CO4	2
Q3.	A dataset has values 4, 8, 12, 16, 20, 24, 28. Construct a box plot by identifying the minimum, Q1, median, Q3, and maximum.	CO3	4
Q4.	Discuss data transformation strategies, define any five strategies in detail.	CO4	5

Section B

S.No	Question	CO	BL
Q1.	Explain the different steps involved in data pre-processing and discuss how each step contributes to improving data quality.	CO2	2
Q2.	A teacher recorded the test scores of 10 students: 55, 60, 65, 70, 70, 75, 80, 85, 90, 95. Calculate the mean, median, mode, and standard deviation.	CO3	3

PAPER CODE: BCE-O633
SEMESTER EXAMINATION (MAY-2025)

CLASS: B.TECH.

SEMESTER: VI

SUBJECT: COMPUTER SCIENCE AND ENGINEERING

PAPER TITLE: INTRODUCTION TO DATA SCIENCE AND DESIGN THINKING

Time: 3 Hours

Max. Marks: 70

Note: Question Paper is divided into two sections: **A and B**. Attempt both the sections as per given instructions.

SECTION-A (SHORT ANSWER TYPE QUESTIONS)

Instructions: Answer any *five* questions in about 150 words each. Each question carries six marks. **(5 x 6 = 30 Marks)**

		CO	BL
Q-1	How does Design Thinking contribute to solving real-world problems? Illustrate with a suitable approach.	1	2
Q-2	What are Skewness and Kurtosis in data analysis, and why are they important for understanding data distribution?	2	3
Q-3	Explain the concept of Data Integration. What obstacles commonly arise during integration in complex data systems?	1	4
Q-4	What makes a design "good"? List key features of a good design and contrast them with traits of a bad design.	3	3
Q-5	Define Data Science. In what ways is it transforming industries and everyday life?	3	2
Q-6	What is a Heat Map? Where are heat maps most effectively used in data analysis or visualization?	2	4
Q-7	In a Data Science project, what roles are involved, and how do they collaboratively drive the project from start to finish?	2	2
Q-8	Discuss five commonly used data collection methods. How do these support effective data-driven decision-making?	1	3

Q-9	What are the five Vs of Data? Explain their relevance in the context of modern data science applications.	3	4
Q-10	How would you apply the principles of design thinking in the creation of an eco-friendly and sustainable product?	4	3
SECTION-B (LONG ANSWER TYPE QUESTIONS) Instructions: Answer any <i>four</i> questions in detail. Each question carries 10 marks. (4 x 10 = Marks)		CO	BL
Q-1	What is Data Science? Explain its applications in at least five different fields with real-world examples.	1	3
Q-2	Describe the stages of a Data Science project. How do these stages align with the Data Analytics Life Cycle?	2	4
Q-3	Discuss data transformation strategies, define any five strategies in detail.	2	3
Q-4	What is Data Integration? Explain the key challenges faced during integration in large-scale data environments.	1	4
Q-5	Calculate the mean, median, and mode of the following dataset: 24, 22, 34, 112, 12, 145, 56, 78, 90, 34, 21, 67, 90, 55, 23, 50, 34 Also, create a box plot to visually represent the data.	3	4
Q-6	Explain the five stages of the Design Thinking process. Provide examples for each stage to support your explanation.	1	3
Q-7	What are the different ways Design Thinking can be applied for Rural Development? Discuss with practical use cases.	3	5
Q-8	Discuss the following Use cases with respect to application of Data Science and Design thinking for problem solving a) Amazon Go store b) The Good Kitchen case study	2	3

Sessional Exam I

Date: 25/02/2026

Time: 2:00- 3:00 PM

Introduction to Data Science and Design Thinking (BCE-O643)

Note: Section A contains 4 questions of 6 marks each attempt any 2.

Section B contains 2 questions of 8 marks each, attempt any 1.

Section A

- Q1.** Define *Data Science*. Explain the step-by-step approach to learning Data Science. (CO1)
- Q2.** Describe the five key characteristics of data and explain their significance. (CO1)
- Q3.** What are the major stages involved in a Data Science project? Briefly explain each phase. (CO2)
- Q4.** Discuss the major data security issues and challenges encountered in a Data Science project. (CO2)

Section B

- Q1.** Suppose you are the owner of a Vehicle Rental Company aiming to increase revenue in the upcoming semester. Propose a data-driven strategy to achieve this objective. Discuss your approach in terms of different Data Science roles and clearly explain the responsibilities of each role/persona involved. (CO1)
- Q2.** Explain the applications of Data Science across various domains. Describe any five applications in detail. (CO1)

Introduction to Data Science and Design Thinking (BCE-O633)

Sessional Exam II

Date: 22/04/2026

Time: 2:00- 3:00 PM

Note: Section A contains 4 questions of 6 marks each attempt any 2.

Section B contains 2 questions of 8 marks each, attempt any 1.

Section A

S.No	Question	CO	BL
Q1.	What is data Collection? Discuss the different methods for primary Data Collection.	CO3	3
Q2.	What is Data Reduction? Discuss any two Data Reduction Strategies.	CO4	2
Q3.	A dataset has values 4, 8, 12, 16, 20, 24, 28. Construct a box plot by identifying the minimum, Q1, median, Q3, and maximum.	CO3	4
Q4.	What is design thinking? define in your words.	CO4	5

Section B

S.No	Question	CO	BL
Q1.	What are the stages in Design Thinking? elaborate each stage with diagram.	CO3	2
Q2.	A teacher recorded the test scores of 10 students: 55, 60, 65, 70, 70, 75, 80, 85, 90, 95. Calculate the mean, median, mode, and standard deviation.	CO3	3

PAPER CODE: BCE-O633
EVEN SEMESTER EXAMINATION (MAY-2026)
CLASS: B.TECH. SEMESTER: VI
PAPER TITLE: INTRODUCTION TO DATA SCIENCE
AND DESIGN THINKING

Time: 3 hours

Max. Marks: 70

Note: Question Paper is divided into two sections: **A and B**. Attempt both the sections as per given instructions.

SECTION-A (SHORT ANSWER TYPE QUESTIONS)

Instructions: Answer any *five* questions in about 150 words each. Each question carries six marks.
 (5 x 6 = 30 Marks)

Q.No.		CO	BL
Q1	Explain the evolution of Data Science and how it differs from traditional statistics.	CO1	BL2
Q2	Define Data Science roles and describe any three roles involved in a data science project.	CO1	BL2
Q3	What is data cleaning? Explain techniques for handling missing data.	CO4	BL2
Q4	Differentiate between data integration and data transformation with examples.	CO4	BL2
Q5	Calculate the Mean Median and Mode for the dataset: 5, 5, 10, 12, 13, 15, 16, 18, 20, 20, 20, 25	CO2	BL3
Q6	What is correlation? Explain positive and negative correlation with examples.	CO2	BL2
Q7	Define design thinking and explain its importance in product development.	CO6	BL2
Q8	Explain the Empathize and Define stages of Design Thinking with examples.	CO7	BL2
Q9	How does design thinking help in rural transport design? Explain with reference to the bullock cart.	CO8	BL3
Q10	Write short notes on (any two): (a) Bicycle as ecological design (b) Toothbrush as cultural design (c) Duster as appropriate technology	CO8	BL2

SECTION-B (LONG ANSWER TYPE QUESTIONS)

Instructions: Answer any *four* questions in detail. Each question carries 10 marks.

(4 x 10 = 40 Marks)

Q.No.		CO	BL
Q1	Explain the complete lifecycle of a Data Science project with a real-world example.	CO1	BL 3
Q2	Discuss various data collection strategies. Compare primary and secondary data collection methods.	CO3	BL 4
Q3	Explain data preprocessing techniques including data reduction and discretization. How do they improve data quality?	CO4	BL 4
Q4	What are data security issues? Write any 5 data security issues in detail.	CO2	BL 3
Q5	The marks obtained by 10 students in a mathematics test are as follows: 45, 50, 55, 50, 60, 65, 70, 50, 75, 80 (a) Find the mean of the given data. (b) Determine the median of the marks. (c) Identify the mode of the given dataset. (d) Compute the range of the data. (e) Find the inter quartile range (f) Draw a box plot for the given data.	CO2	BL 4
Q6	Describe all stages of Design Thinking with a practical case illustration.	CO6	BL 3
Q7	Discuss a case study of Tesla or AIRBNB explaining the role of design thinking in innovation.	CO9	BL 4
Q8	Discuss the principles of Good Design and Bad Design. Elaborate with the help of examples.	CO8	BL 4
